

CSC1060 - Data Analytics

Fundamentals Problem Solving 1 -

Data Analytics in Healthcare

Sample Document for student reference

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Introduction

Healthcare organisations use data analytics to monitor and improve results for patients and support decision making. With large data available, it gives researchers the ability to identify disease patterns and recognise vulnerable population groups. One chronic condition that effects older adults is arthritis. Arthritis is term for many conditions that affect the joints of a body, it is medical condition that is only common amongst people aged in their late 40s or above. Its effects include pain, reduced mobility, partial disability and a toll on quality of life making it an issue for public health.

This report provides the data analysis needed to explore arthritis among older adults using the Centres for Disease Control and Prevention (CDC), Alzheimer's Disease and Healthy Aging dataset. Python was used to prepare, clean, analyse and visualise the data. Statistical analysis and a predictive model were then applied to better understand arthritis cases across different demographic groups and geographic locations. The goal of this report is to identify which groups are most affected by arthritis and to see whether a predictive model can estimate arthritis occurrence based on historical data. The results aim to provide healthcare professionals with useful data so that it may assist future planning and preventative programs.



Problem Definition

2.1 Background and Context

According to the Centres for Disease Control and Prevention (CDC), Arthritis is one of the leading causes of chronic pain and disability older adults experience. Arthritis affects a large population of adults and often has a toll on performing physical activity and therefore reduces independence, this then contributes to other chronic health conditions.

Osteoarthritis is one of the most common chronic conditions affecting older adults and is associated with pain, reduced mobility and decreased quality of life. As populations continue to age, the burden of arthritis is expected to increase, creating additional pressure on healthcare systems and community support services (French et al., 2016).

Understanding which demographic groups and which locations experience higher arthritis cases allows healthcare providers to assign resources more effectively and develop prevention strategies. Data analytics allows us to identify the patterns by analysing large-scale population health datasets.

2.2 Business Case and Objectives

Healthcare organisations require reliable evidence so that they make decisions such as funding and prevention programs. Without analysing the population data, the targeted resource will almost be unlikely to go towards the groups that need them most.

Based on the Alzheimer Disease and Healthy Aging Data provided by the Centres for Disease Control and Prevention (CDC), we can identify demographic and geographic patterns associated with arthritis cases among older adults. By analysing arthritis cases across different locations, sexes, race/ethnicity groups, the project can be able to provide insights that support healthcare planning and specific treatment strategies.

2.3 Research and Discovery

Before reviewing the dataset, we explored further to better understand arthritis and its impact on ageing. Earlier research shows that common cases of arthritis differ across age groups, sex, ethnicity, lifestyle factors and geographic regions in the United States. Older adults that live with the condition experience reduced mobility, have higher healthcare consumption and unpleasant quality of life. The dataset provides a helpful source of the public health information because it contains demographic and geographic statistics collected across the United States. Looking at these variables allows researchers to recognise patterns that support healthcare projections.



Data Understanding

3.1 Data Description

The dataset used in this report was taken from the CDC Alzheimer's Disease and Healthy Aging Data collection. The dataset contains information on many health signs affecting older adults all over the United States. We only analysed the records of the topic column "Arthritis among older adults" and after filtrations we produced a dataset of 6,053 cases. The variables that were selected on analyses included:

- Arthritis occurrence
- Location
- Year
- Age Group
- Sex
- Race/Ethnicity

The chosen variables were due to directly support the research question by allowing comparisons across demographic and geographic groups.

3.2 Initial Data Exploration

The initial exploration was viewed using Python and its built-in the pandas function. The dataset managed to identify factors such as missing values, variable types, summary statistics and overall distribution of arthritis cases.

Initial patterns from exploring this data has indicated average arthritis cases of approximately 42.3%, with values ranging from 9.5% to 77.5%. This big range indicates a large variation between different locations and demographic groups. It has also showed that the dataset contained information for several years, multiple states and several demographic classes such as sex and ethnicity. With these variables, they were then considered suitable for further analysis and visualisation.

```
count    6053.000000
mean      42.285313
std       9.784008
min       9.500000
25%      35.400000
50%      42.300000
75%      49.400000
max      77.500000
```

The interquartile range also provides useful information, in the first quarter of dataset had cases below 35.4% while the third quarter had cases below 49.4% which indicates that most cases were mainly between 35% and 49% approximately.

```
Name: Data_Value, dtype: float64
```

Data Preparation

4.1 Data Cleaning

The first step in the data cleansing was to import the CSV dataset into Python using the Pandas function. Then dataset was inspected to understand its structure, identify the variables and spot any missing values that could affect the quality of the analysis.

The dataset was then filtered to include only records of "Arthritis among older adults." to ensure the analysis remained focused on the purpose of the report. By implementing this step, it removed unrelated health topics and produced a dataset specifically relevant to arthritis cases.

The dataset was then observed for missing values using 'isnull().sum()' Python function. The "Data_Value" column contained 2,250 missing values, however several other variables including "Data_Value_Alt", "Low_Confidence_Limit" and "High_Confidence_Limit" also showed missing data.

Since "Data_Value" represents the percentage of older adults diagnosed with arthritis and was the primary variable used throughout the analysis, all data with missing "Data_Value" values were removed using 'dropna()' function. By doing this, it ensured that statistics analysis, visualisations and the predictive model were created on complete observations which will reduce the chances of inaccurate results therefore improving the reliability of the analysis.

```
RowId 0
YearStart 0
YearEnd 0
LocationAbbr 0
LocationDesc 0
Datasource 0
Class 0
Topic 0
Question 0
Data_Value_Unit 0
DataValueTypeID 0
Data_Value_Type 0
Data_Value 2250
Data_Value_Alt 2250
Data_Value_Footnote_Symbol 5952
Data_Value_Footnote 5952
Low_Confidence_Limit 2250
High_Confidence_Limit 2250
StratificationCategory1 0
Stratification1 0
StratificationCategory2 1044
Stratification2 1044
Geolocation 720
ClassID 0
TopicID 0
QuestionID 0
LocationID 0
StratificationCategoryID1 0
StratificationID1 0
StratificationCategoryID2 0
StratificationID2 0
dtype: int64
```

4.2 Data Transformation

Once the steps of data cleansing were concluded, the dataset was changed into a format which was appropriate for analysis. Using the appropriate commands, the data was divided the groups into the following:

- Average arthritis cases by state
- Average arthritis cases by sex
- Average arthritis cases by race/ethnicity
- Average arthritis cases by year

By dividing this dataset, it transformed datasets which were then visualised using bar and line graphs to identify important patterns and differences between demographic groups.

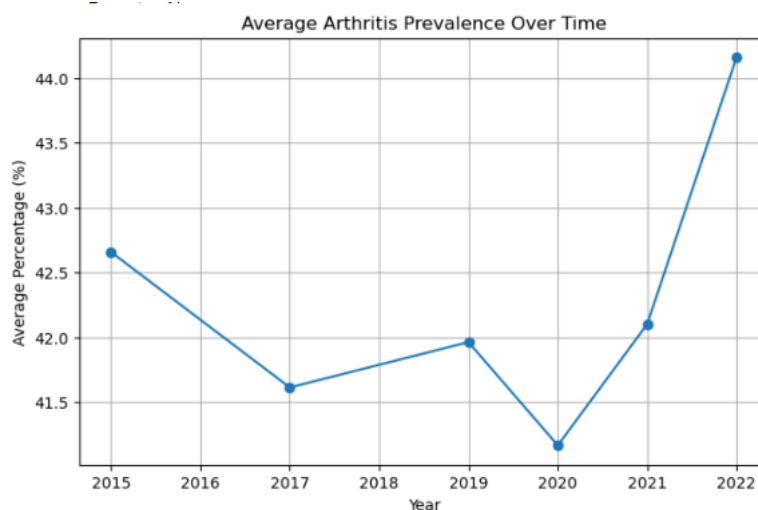
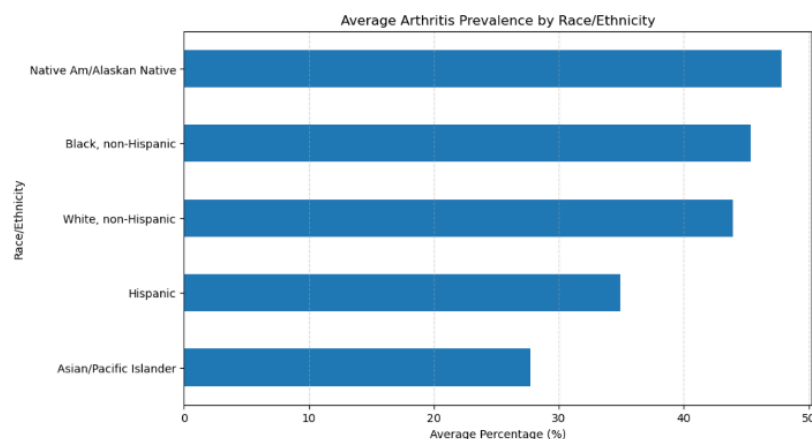
	LocationDesc	YearStart	Data_Value	StratificationCategory1	Stratification1	StratificationCategory2	Stratification2
2	Midwest	2015	32.0	Age Group	50-64 years	Sex	Male
4	Missouri	2015	46.8	Age Group	Overall	NaN	NaN
8	South	2015	36.8	Age Group	50-64 years	NaN	NaN
9	Tennessee	2015	47.3	Age Group	Overall	Sex	Male
15	Michigan	2015	46.4	Age Group	Overall	Race/Ethnicity	Black, non-Hispanic

Data Analysis and Visualisation

Using Python, data analysis and visualisation were performed to find patterns in arthritis cases among older adults. After cleaning the dataset and removing records with missing values, the dataset contained 6,053 cases. With an average arthritis case of 42.29%, a minimum value of 9.5% and a maximum value of 77.5%, showing a strong variation across the population. It also showed a standard deviation of 9.78, suggesting noticeable patterns between geographic locations and demographic groups.

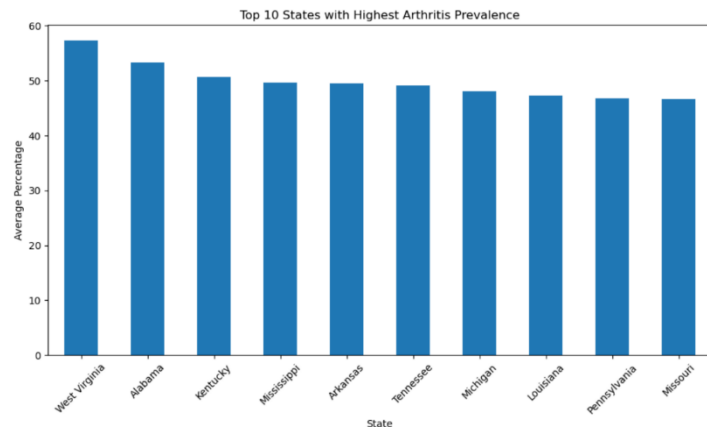
The findings of this study emphasise previous research demonstrating that arthritis remains a significant public health concern among older adults. Data analytics can assist healthcare organisations in identifying high-risk populations and supporting evidence-based decision making for prevention and intervention strategies (French et al., 2016).

Gomez et al. (2025) reported that osteoarthritis prevalence increased with age and was significantly higher among females than males. The study also found strong associations between osteoarthritis and obesity, multimorbidity, mobility disability, poor self-rated health and reduced physical activity. These findings support the importance of examining demographic and health-related factors when analysing arthritis cases among older adults.



5.1 Arthritis Prevalence by Location

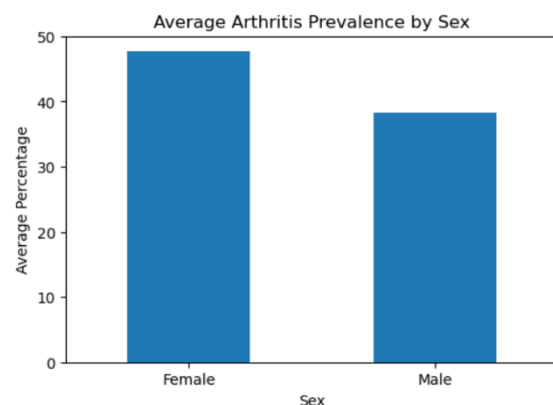
The first analysis examined the average arthritis cases across different locations. The results showed that West Virginia had the highest average cases of arthritis among older adults with around 57.3%, followed by Alabama with around 53.3%, Kentucky came third with around 50.7%, fourth came Mississippi with around 49.7% and fifth was Arkansas with around 49.6%.



The differences between locations may be influenced by factors such as age distribution, socioeconomic conditions, healthcare access, obesity, physical activity levels as well as occupational factors. Identifying states with higher cases allows healthcare organisations to prioritise preventative programs and distribute healthcare resources accordingly.

5.2 Arthritis Prevalence by Sex

The second analysis made comparison between males and females with arthritis cases. The results shows that females had a higher average of arthritis cases with around 47.6% where males were only 38.2%.



This pattern is consistent with previous research portraying that women are more likely to develop arthritis, particularly osteoarthritis. Due to hormone changes, longer life expectancy as well as biological differences it may impact to the increased cases with females. These findings could imply that prevention and management programs may need to consider sex specific factors when creating public health programs.

References

French, H. P., Galvin, R., Horgan, N. F., & Kenny, R. A. (2016). Prevalence and burden of osteoarthritis amongst older people in Ireland: Findings from the Irish Longitudinal Study on Ageing (TILDA). *European Journal of Public Health*, 26(1), 192–198.

Gomez, F., Duque, G., Curcio, C.-L., Cano, L., Osorio, D., & Rios, S. (2025). *Prevalence and burden of osteoarthritis amongst older people in Colombia: Results from national survey of health, wellbeing, and aging (SABE Colombia)*. *Clinical Epidemiology and Global Health*, 31, 101829.

[https://www.google.com/search?q=Centres+for+Disease+Control+and+Prevention+\(CDC\),&sca_esv=d81a8aeecb3ef84&rlz=1C1CHBF_en-GBAU1105AU1105&sxsrf=APpeQnuNo13aGqub8kMg0gn2CPz09atAeQ:1783469913664&udm=2&fbs=ABfTbFVyMZGZf1hfvX9uKjN_-G8cY2oODYyTyZk24Xz37_7FQ0Kuh3jfnRqiKzI7KrGsO-bHLKGXxP60AsAIKmX_LR6d5sFoO4fEienlZG8AK4GBiPIIR-Wi0jE5Gc6lfyzKs_n5ApRmsnDMsNUTOEGDxGAazg0-zfc66RhCrRLtnQp3KB-A5JZsMBV3DI-4Km7YR2eujYSQSGQ_PdZCX-2vskDL6dluOg&ved=2ahUKEwivK-V58GVAxXUTGwGHXw1AAUQtKgLegQIAxAH&biw=1280&bih=665&dpr=2#sv=CAMSURoyKhBILVZPb01FTzFDMVJWaHJNMg5WT29NRU8xQzFSVmhyTToOWUWtZa2dk11RGw4YU0gBCoXCgFzEhBILVZPb01FTzFDMVJWaHJNGAEwARgHII6ZjrkPSggQARgBIAEoAQ](https://www.google.com/search?q=Centres+for+Disease+Control+and+Prevention+(CDC),&sca_esv=d81a8aeecb3ef84&rlz=1C1CHBF_en-GBAU1105AU1105&sxsrf=APpeQnuNo13aGqub8kMg0gn2CPz09atAeQ:1783469913664&udm=2&fbs=ABfTbFVyMZGZf1hfvX9uKjN_-G8cY2oODYyTyZk24Xz37_7FQ0Kuh3jfnRqiKzI7KrGsO-bHLKGXxP60AsAIKmX_LR6d5sFoO4fEienlZG8AK4GBiPIIR-Wi0jE5Gc6lfyzKs_n5ApRmsnDMsNUTOEGDxGAazg0-zfc66RhCrRLtnQp3KB-A5JZsMBV3DI-4Km7YR2eujYSQSGQ_PdZCX-2vskDL6dluOg&ved=2ahUKEwivK-V58GVAxXUTGwGHXw1AAUQtKgLegQIAxAH&biw=1280&bih=665&dpr=2#sv=CAMSURoyKhBILVZPb01FTzFDMVJWaHJNMg5WT29NRU8xQzFSVmhyTToOWUWtZa2dk11RGw4YU0gBCoXCgFzEhBILVZPb01FTzFDMVJWaHJNGAEwARgHII6ZjrkPSggQARgBIAEoAQ)

https://www.google.com/search?q=athritis++usa&sca_esv=d81a8aeecb3ef84&rlz=1C1CHBF_en-GBAU1105AU1105&udm=2&biw=1280&bih=665&sxsrf=APpeQntORFgfBO9sf4Nd_Xx3jFMNgWuXIq%3A1783471447018&ei=V51Naqdg-Y6d6w-z-bjACg&ved=0ahUKEwjn58Pw7MGVAXV5R2cHHbM8DqgQ4dUDCB&uact=5&oq=athritis++usa&gs_lp=Egtnd3Mtd2l6LWltZylnYXRocml0aXMgIHVzYUj9BFDWA1jWA3ABeACQAQCQAQCgAQcQAQC4AQPIAQD4AQGYAgCgAgCYAwDiAwUSATEgQlGGAZIHAKAHALgHAMIHAMgHALIAAQ&sclient=gws-wiz-img#sv=CAMSURoyKhBILVdNbGx0SEozOGpXOFBNMg5XTWxsdEhKMzhqVzhQTToOeXc4dGJ5dXINX0xfSU0gBCoXCgFzEhBILVdNbGx0SEozOGpXOFBNGAEwARgHIOrntqgMSggQARgBIAEoAQ